

THE GENESIS PROTOCOL 2.0: GLOBAL VALUE CAPTURE

Building the Operating System for Physical Matter

THE VISION

We are not a battery company. We are not even a materials company.

We are building the AI infrastructure layer for the physical world.

Just as AWS became the operating system for digital products, **Skanda becomes the operating system for physical products**. Every advanced manufacturer—from EV batteries to hydrogen fuel cells to low-carbon cement—will run their designs through our platform before building anything.

The Thesis:

- Google/Microsoft operate at the atomic level (DFT) → Too small to predict factory yield
- Traditional FEA operates at the system level → Too big to understand material failure
- **We own the mesoscale** (10nm-100µm) → Where 90% of industrial materials actually fail

The Opportunity:

- \$300B battery market (our beachhead)
- \$8-10T in materials across hydrogen, cement, alloys, semiconductors, chemicals
- Zero credible competitors at the mesoscale (18-month head start)

The Outcome:

- 18 months: \$3-5B valuation (platform proven, \$50-60M ARR)
- 5 years: \$40-60B valuation (IPO or strategic acquisition)
- 10 years: The defining AI infrastructure company for the physical world

THE PLATFORM ARCHITECTURE

The Foundation: Physics, Not Just Pattern Matching

What we built:

- **SkandaX Foundation Model:** Trained on 10M+ synthetic physics simulations spanning the entire "Physics Hypercube" (diffusion, kinetics, thermodynamics)
- **Real-World Calibration:** Fine-tuned on 1,000+ multimodal samples (recipe + microstructure + performance)
- **Transfer Learning:** Each new chemistry requires 70% less data than the previous one

Why this is defensible:

- Competitors can't just "train on more data" (real mesoscale data doesn't exist at scale)
- We manufactured our own training distribution using governing physics equations
- 10M simulations took 12 months + deep expertise (not reproducible quickly)

The Technical Moat:

Physics Intuition (10M synthetic samples)

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Foundation Model (universal transport laws)

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Real-World Calibration (1,000+ samples)

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Transfer Learning (new chemistry in <100 samples)

THE REVENUE ENGINE: FOUR COMPOUNDING LAYERS

We capture value across four scalable layers, each with different economics and growth rates.

LAYER 0: THE API PLATFORM (Primary Growth Engine)

Revenue Stream: Usage-based pricing + Self-service subscriptions

The Product: SkandaX API — developers query the physics model directly

How It Works:

POST /predict-performance
Input: Recipe + Microstructure image
Output: Cycle life, failure mode, confidence
Price: \$0.01 per prediction

POST /generate-design
Input: Target specs (cycle life, power, temperature)
Output: 3D microstructure + manufacturing recipe
Price: \$1.00 per generation

POST /optimize-recipe
Input: Current recipe + performance gap
Output: Optimized parameters
Price: \$0.50 per optimization

Pricing Tiers:

TIER	MONTHLY FEE	INCLUDED USAGE	OVERAGE RATE	TARGET CUSTOMER
Free	\$0	100 predictions	N/A	Researchers, students
Developer	\$500	10,000 predictions	\$0.01/prediction	Startups, labs
Professional	\$5,000	100,000 predictions	\$0.008/prediction	SMB manufacturers
Enterprise	\$50-500k	Unlimited	Custom SLA	OEMs, gigafactories

The Scaling Math:

Month 12:

- 50 Professional customers × \$5,000/month = \$250k/month

- 200 Developer customers × \$500/month = \$100k/month
- Overage revenue (20% of customers) = \$50k/month
- **Total: \$400k/month = \$4.8M ARR**

Month 18:

- 200 Professional × \$5,000 = \$1M/month
- 500 Developer × \$500 = \$250k/month
- 10 Enterprise × \$100k = \$1M/month
- Overage = \$250k/month
- **Total: \$2.5M/month = \$30M ARR**

Month 24:

- 500 Professional × \$5,000 = \$2.5M/month
- 2,000 Developer × \$500 = \$1M/month
- 50 Enterprise × \$150k = \$7.5M/month
- Overage = \$1M/month
- **Total: \$12M/month = \$144M ARR**

Why This Scales Exponentially:

-  **Viral adoption:** Researchers share, startups upgrade to paid
-  **Network effects:** More usage → better model → more customers
-  **Zero marginal cost:** 90%+ gross margins (pure software)
-  **Global distribution:** No sales team needed for <\$50k deals

The Comparable: This is the Databricks/Snowflake model. Platform usage compounds.

Investor Note: By Month 24, API alone could be worth \$2-4B at 20-30x ARR multiples.

LAYER 1: STRATEGIC DESIGN PARTNERSHIPS (The Bridge)

Revenue Stream: High-margin NRE fees + IP co-creation

The Model: Partners pay \$3-7M for exclusive access to our generative engine for 18 months

What We Deliver:

- Forward Deployed Engineers (FDEs) embedded at partner R&D center
- Custom chemistry design using SkandaX GENESIS
- Proprietary microstructure IP (co-owned or exclusively licensed)
- Manufacturing recipe + process parameters

The Structure:

COMPONENT	PARTNER PAYS	WE PROVIDE	IP OWNERSHIP
Discovery Phase (6 months)	\$2M upfront	2-3 FDEs, unlimited API access	Joint patent filing
Development Phase (6 months)	\$2M milestone	Lab validation, recipe optimization	We retain platform rights
Deployment Phase (6 months)	\$1-3M milestone	Factory integration, yield optimization	They get 3-5 year exclusivity

Key Innovation: After exclusivity expires, we can license the IP to others OR offer via API

Example Partnership Flow:

BMW pays \$5M for next-gen Silicon anode:

1. Month 0-6: SkandaX generates 20 candidate microstructures → BMW tests 5 in lab → 1 works
2. Month 6-12: Optimize winning design for BMW's specific factory equipment
3. Month 12-18: Deploy "Factory Guard" software at BMW pilot line
4. **Outcome:** BMW gets exclusive Silicon IP until 2028, we get \$5M + learnings that improve platform

After 2028: We can license the same chemistry to VW, Tata, etc. at \$1-2M each

Target Customers:

- **Tier-1 OEMs:** BMW, Tesla, Tata (proprietary chemistries)
- **Battery Startups:** QuantumScape, Factorial (co-development)
- **Defense/Aerospace:** Lockheed, Northrop (extreme-condition batteries)

The Scaling Math:

Year 1 (Months 0-12):

- 3-5 partnerships × \$4M average = **\$12-20M**
- Focus: Batteries only (prove the model)

Year 2 (Months 12-24):

- 8-12 partnerships × \$5M average = **\$40-60M**
- Expand: Hydrogen, cement, alloys

Year 3+:

- 15-20 partnerships/year × \$5M = **\$75-100M/year**
- Plus: License expired IPs to secondary customers

Why This Matters:

-  **Non-dilutive capital:** Funds operations without raising equity
-  **Data moat:** Every partnership generates proprietary samples
-  **Proof points:** "BMW chose us" = validation for others
-  **IP accumulation:** Each deal creates an asset we monetize forever

Gross Margin: 60-70% (mostly labor, some lab costs)

LAYER 2: THE YIELD OPERATING SYSTEM (Recurring SaaS)

Revenue Stream: Annual subscriptions per production line + performance bonuses

The Product: SkandaX DEPLOY — Real-time factory monitoring and defect prediction

How It Works:**Step 1: Integration**

- Deploy edge node (NVIDIA Orin/IGX) directly on factory floor
- Connect to production line sensors (coating speed, humidity, temperature)
- Federated learning: Data never leaves factory, only gradients sent to central model

Step 2: Real-Time Monitoring

- Camera captures electrode images during coating
- AI detects micro-defects invisible to human QC inspectors
- Flags "bad batches" before they reach cell assembly

Step 3: Yield Optimization

- Alerts operators to process drift ("Humidity spiked, increase drying time by 8%")
- Predicts failures before they happen ("This batch will have 15% higher resistance")
- Recommends corrective actions in real-time

The Pricing Model:

Base Subscription:

- \$250k-500k per production line per year
- Includes software, updates, support
- 3-year minimum contract

Performance Bonus:

- 10% of documented savings (yield improvement)
- Example: Factory reduces scrap from 15% → 8% = \$12M saved
- Our cut: \$250k base + \$1.2M bonus = **\$1.45M total**

Equity Kicker (Optional):

- For discounted pricing, take small warrant (0.1-0.5%)
- Aligns incentives: we win when customer scales production

Target Customers:

CUSTOMER TYPE	LINES	PRICE	ANNUAL REVENUE
Gigafactory (CATL, LG)	20-50 lines	\$500k/line	\$10-25M
Mid-Tier (Northvolt, Tata)	5-10 lines	\$350k/line	\$1.75-3.5M
Startups (Our NRE partners)	1-2 lines	\$250k/line	\$250-500k

The Scaling Math:

Year 1 (Months 6-12):

- 5-10 production lines deployed
- Average \$300k/line (early adopter discount)
- **Revenue: \$1.5-3M ARR**

Year 2 (Months 12-24):

- 30-50 production lines
- Average \$400k/line (pricing power from results)
- **Revenue: \$12-20M ARR**

Year 3+:

- 100+ production lines globally
- Average \$500k/line + performance bonuses
- **Revenue: \$60-100M ARR**

Why This Is Sticky:

-  **Switching costs:** Once integrated into PLC/MES, cannot be removed
-  **Improves over time:** Federated learning makes it smarter with every batch
-  **ROI is clear:** Prevents one bad batch = 10x the annual fee

Gross Margin: 85-90% (software + edge hardware, minimal ongoing costs)

The Lock-In: This is the "Operating System" play. Once we're in the factory stack, we become infrastructure.

LAYER 3: IP ROYALTIES (The Exponential Upside)

Revenue Stream: Production royalties on Skanda-designed chemistries

The Model: \$0.50-1.50 per kWh produced using our proprietary recipes

How It Works:

Scenario A: Co-Developed IP (from Layer 1 partnerships)

- We design proprietary microstructure for BMW's Silicon anode
- BMW has 3-year exclusivity (no royalty during exclusivity)

- After 2028: BMW pays \$0.75/kWh OR we license to others at \$1.25/kWh

Scenario B: Platform-Generated IP (from Layer 0 API)

- Startup uses our API to generate novel Sodium-ion design
- They commercialize it → royalty kicks in when production >10 GWh/year
- Rate: \$0.50/kWh (lower because they used self-service)

Scenario C: Owned IP (our R&D generated)

- We discover breakthrough solid-state electrolyte microstructure
- License non-exclusively to multiple manufacturers
- Rate: \$1.50/kWh (highest because we own 100% of IP)

The Unit Economics:

Conservative Scenario (Year 5):

- We have IP on 10% of next-gen battery production
- Global next-gen capacity: 200 GWh/year
- Our share: 20 GWh × \$1/kWh average = **\$20M/year**

Aggressive Scenario (Year 7):

- We have IP on 25% of next-gen chemistries
- Global capacity: 500 GWh/year
- Our share: 125 GWh × \$1.25/kWh = **\$156M/year**

Bull Case (Year 10):

- We dominate Silicon, Sodium, Solid-State IP
- 30% of global advanced battery production
- 1,000 GWh/year × 30% × \$1/kWh = **\$300M/year**

Why This Is The "ARM Play":

-  **Exponential scaling:** Revenue grows with industry, not our headcount
-  **Zero marginal cost:** Pure profit after IP creation
-  **Compounding:** Every chemistry we solve → permanent revenue stream
-  **Defensible:** Patents + trade secrets create 15-20 year moat

Gross Margin: 95%+ (no COGS, just IP licensing)

Valuation Impact: At 30x multiples, \$300M royalty revenue = **\$9B in valuation from IP alone**

THE 18-MONTH EXECUTION PLAN

How we go from proof-of-concept to platform dominance in 18 months.

MONTH 0-6: THE TRIPLE SPRINT

We execute three workstreams in **PARALLEL**, not sequentially.

SPRINT A: PROVE THE SCIENCE (SILICON VALIDATION)

Goal: Achieve >90% R^2 on real-world cycle life prediction

Execution:

- Generate 300-500 Silicon anode samples using Parent-Child protocol
- Multimodal data: Recipe + SEM images + XRD patterns + cycling curves
- Fine-tune forward model on real data (currently 99% R^2 on synthetic, target 90%+ on real)
- **Critical milestone:** Generate one AI-designed recipe → Build in lab → Achieves target performance

Deliverables:

-  Forward model validated on real factory-like data
-  Inverse model generates manufacturable designs (80% success rate)
-  One "zero-shot" recipe works in lab (proves the loop closes)
-  Paper submitted to Nature Materials or Science

Team: 2-3 lab technicians, 2-3 ML engineers, 1 materials scientist

Budget: \$500k (equipment, materials, salaries)

SPRINT B: LAUNCH API BETA (PRODUCT VALIDATION)

Goal: Prove product-market fit for API platform

Execution:

- Ship MVP API with forward model only (prediction, not generation yet)
- Target: 50 beta users (universities, startups, R&D labs)
- Pricing: Free tier for researchers, \$0.01/prediction for commercial
- Collect feedback, iterate weekly

Deliverables:

-  API live at api.skanda.ai with 99% uptime
-  50+ registered users generating 10,000+ predictions/month
-  10-15 paying customers validating willingness-to-pay
-  Net Promoter Score >50 (product-market fit signal)

Team: 3-4 platform engineers, 1 developer advocate, 1 designer

Budget: \$300k (eng salaries, cloud infrastructure)

SPRINT C: SECURE ANCHOR CUSTOMERS (REVENUE VALIDATION)

Goal: Close \$15-25M in NRE partnerships

Execution:

- Target 10-15 enterprise prospects (OEMs, Tier-1 suppliers, defense)
- Pitch: "We'll design your next-gen chemistry in 6 months instead of 3 years"
- Close 3-5 deals at \$3-5M each
- Deploy FDEs at partner sites

Deliverables:

-  \$15-25M in signed contracts (non-dilutive capital)
-  3-5 active partnerships (BMW/Tesla/Tata/QuantumScape tier)

-  5-8 FDEs deployed globally

Team: 2-3 enterprise sales, 1 partnerships lead, 5-8 FDEs

Budget: \$800k (sales team, travel, FDE salaries for 6 months)

END OF MONTH 6 OUTCOME:

METRIC	TARGET	WHY IT MATTERS
Scientific	R ² >90% on real data	Proves AI works, not just on synthetic
Product	50+ API users, 10 paying	Validates platform business model
Revenue	\$15-25M contracts signed	Non-dilutive funding for next phase
Valuation	\$300-500M	Technical + commercial de-risking

MONTH 6-12: THE PLATFORM LAUNCH

Mission: Ship production platform, expand chemistry coverage, cross \$50M cumulative revenue

TECHNICAL EXPANSION (THE FOUNDATION MODEL THESIS)

Add 3-4 New Chemistries:

CHEMISTRY	WHY	SAMPLE TARGET	TRANSFER LEARNING EFFICIENCY
NMC Cathode	Pairs with Silicon, high EV demand	150 samples	70% less data (same physics)
LFP Cathode	Easier, validates generalization	100 samples	75% less data
Hydrogen PEM	New market, \$200B TAM	150 samples	70% less data (transfer learning)

Sodium-Ion	Emerging market, China/India focus	150 samples	70% less data
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The Validation:

- If NMC model achieves 85%+ R^2 with only 150 samples (vs Silicon's 300)
- **We've proven transfer learning works** → Foundation model is real
- This unlocks every future chemistry at 70% lower cost

Expected Outcome:

-  Foundation Model v2 trained (unified electrochemistry brain)
-  4 chemistries validated (Silicon, NMC, LFP, Hydrogen)
-  Platform can onboard new chemistry in 3-4 months (vs 12 months for first)

PRODUCT LAUNCH (GENERAL AVAILABILITY)

SkandaX Platform v1.0:

API (Production-Ready):

- Forward model (prediction)
- Inverse model (generation)
- Process model (recipe → manufacturing params)
- 99.9% uptime SLA
- SDKs: Python, R, Julia

Dashboard (Self-Service):

- Upload SEM images → Get predictions
- Input target specs → Get designs
- Track simulation history
- Download results (TIFF, JSON, PDF reports)

Documentation:

- API reference
- Tutorial notebooks

- Case studies
- Webinars/workshops

Target: 100-200 active customers by Month 12

COMMERCIAL EXECUTION

API Revenue (Primary Growth):

- 150 Professional tier × \$5,000/month = \$750k/month
- 300 Developer tier × \$500/month = \$150k/month
- 5 Enterprise tier × \$100k/month = \$500k/month
- Overage revenue = \$100k/month
- **Total API: \$1.5M/month = \$18M ARR**

NRE Partnerships:

- 5-7 new deals signed (Year 1 total: 8-12 partnerships)
- Average deal size: \$5M
- **Revenue: \$25-35M (cash collected)**

Factory SaaS:

- Deploy at 15-20 production lines
- Average: \$300k/line/year
- **Revenue: \$4.5-6M ARR**

Total Month 12 Revenue: \$50-60M

GLOBAL EXPANSION

Open International Offices:

- **Munich, Germany** (Europe hub)
 - 2-3 FDEs for BMW, Northvolt, BASF
 - Sales engineer for EU market

- **Tokyo, Japan** (Asia hub)
 - 2-3 FDEs for Panasonic, Murata
 - Partnership lead for Asia

Why This Matters:

- Manufacturing is local (customers want on-site support)
- Time zones matter (24/7 coverage for API)
- Regulatory compliance (EU data sovereignty, Japan quality standards)

END OF MONTH 12 OUTCOME:

METRIC	TARGET	SIGNIFICANCE
Platform	100-200 customers	Product-market fit proven at scale
Revenue	\$50-60M ARR	Puts us in top 1% of SaaS startups
Chemistry Coverage	4 validated	Platform thesis proven
Valuation	\$1-2B	20-30x ARR (AI infra multiples)

MONTH 12-18: THE INDUSTRIAL INTEGRATION

Mission: Become mission-critical infrastructure for advanced manufacturing

TECHNICAL MATURITY

Full-Stack Platform:

- **Digital Twin:** Full-cell simulation (anode + cathode + electrolyte)
- **Federated Learning:** 30+ edge nodes deployed globally
- **Multi-Industry:** Expand beyond batteries

New Domains:

INDUSTRY	APPLICATION	WHY NOW	EXPECTED IMPACT
Cement	Low-carbon formulations	Climate mandates, huge TAM	Add \$600B TAM
Aerospace Alloys	Lightweight composites	Defense applications	Add \$200B TAM
Semiconductors	Advanced packaging materials	AI chip demand	Add \$150B TAM

The Thesis:

- Our foundation model understands transport + mechanics + thermodynamics
- These physics apply to cement pore networks, alloy microstructures, chip packaging
- **Batteries were the beachhead, not the endgame**

PRODUCT MATURITY

SkandaX Platform v2.0:

New Products:

- 1. Factory Guard Pro** (enhanced real-time QC)
 - Computer vision + physics model
 - Predicts failures 50-100 cycles before they happen
 - Auto-adjusts process parameters
- 2. Genesis Studio** (no-code inverse design)
 - Drag-and-drop interface for non-engineers
 - "I need X performance" → AI generates options
 - Integrated with CAD tools (export to COMSOL, ANSYS)
- 3. Data Marketplace** (federated learning platform)
 - Partners contribute anonymized data
 - Get credits for API usage

- Platform improves for everyone

Usage Scaling:

- 1,000+ total customers
 - 50+ enterprise contracts (\$500k-2M/year each)
 - Self-service accounts = 80% of volume (20% of revenue)
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COMMERCIAL EXECUTION**API Revenue:**

- 500+ customers × average \$6k/month = \$3M/month
- **API ARR: \$35-45M**

Factory SaaS:

- 25-30 production lines × \$400k/year
- **SaaS ARR: \$10-12M**

IP Royalties:

- First royalty agreements scaling
- **IP ARR: \$3-5M**

NRE Partnerships (Cumulative):

- 15 partnerships × \$4-5M avg = \$60-75M
- **Cumulative NRE Revenue: \$60-75M**

Total Month 18: \$110-135M cumulative (\$50-60M ARR)

STRATEGIC MILESTONES**Operational:**

-  Embedded in 30+ gigafactories globally
-  Platform processes 1M+ simulations/month

-  Largest proprietary mesoscale dataset in the world (10,000+ samples)

Market Position:

-  #1 platform for battery material design
-  First platform expanding to multi-industry
-  Recognized as "the Databricks of materials"

Team:

- 80-120 employees
- 40-50 engineers (ML, platform, data)
- 15-20 FDEs (embedded at customers)
- 10-15 sales/partnerships
- 10-15 lab scientists

END OF MONTH 18 OUTCOME:

METRIC	TARGET	VALUATION DRIVER
Revenue	\$110-135M cumulative (\$50-60M ARR)	Core valuation metric
Customers	500+ total, 15+ enterprise partnerships	Platform adoption
Gross Margin	85-90%	Software economics
Growth Rate	200-300% YoY	Hypergrowth trajectory
Valuation	\$3-5B	50-80x ARR (AI infra multiple)

THE VALUATION TRAJECTORY

How we go from \$0 to \$40-60B in 5 years

The Comparable Companies

COMPANY	INDUSTRY	ARR AT 18MO	VALUATION	MULTIPLE
Databricks	Data infrastructure	~\$200M	\$6B	30x
Snowflake	Data warehouse	~\$100M	\$4B	40x
Scale AI	ML infrastructure	~\$100M	\$3.5B	35x
HashiCorp	DevOps infrastructure	~\$150M	\$5B	33x
Skanda (Projected)	Materials infrastructure	\$50-60M	\$3-5B	50-80x

Why We Command Premium Multiples

1. AI Infrastructure (Not Just Software)

- We're not a SaaS tool, we're the platform layer
- Network effects compound (more data → better model → more customers)
- Winner-take-most dynamics (like AWS, Databricks)

2. Defensible Moat

- **Data:** 10,000+ proprietary multimodal samples (no one else has this)
- **Physics:** Foundation model trained on 10M+ simulations (12-month head start)
- **IP:** 100+ patents on microstructure designs and process recipes

3. Massive TAM

- \$8-10T addressable across materials industries
- Energy transition = secular tailwind (climate mandates)
- Strategic importance (supply chain sovereignty, national security)

4. Capital Efficiency

- 90%+ gross margins (software economics)
- \$20M achieving \$100M+ Silicon Valley equivalent output
- Non-dilutive revenue (NRE partnerships fund growth)

5. Execution Velocity

- AI speed (18-month cycles) not industrial speed (5-year cycles)
 - Parallel expansion (batteries + hydrogen + cement simultaneously)
 - Platform leverage (each chemistry solved makes next easier)
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The 5-Year Path to \$40-60B

Year 1 (Month 18):

- Revenue: \$50-60M ARR (\$110-135M cumulative)
- Valuation: \$3-5B
- Multiple: 50-80x ARR (AI infrastructure premium)
- **Milestone:** Series B at \$3-5B

Year 2:

- Revenue: \$200-300M ARR (4-5x growth)
- Valuation: \$8-12B
- Multiple: 30-40x
- **Milestone:** Series C (optional — may be cash-flow positive)

Year 3:

- Revenue: \$500M-1B ARR (2.5-3x growth)
- Valuation: \$15-25B
- Multiple: 25-30x
- **Milestone:** Pre-IPO round or stay private

Year 4:

- Revenue: \$1.5-2B ARR (2x growth)
- Valuation: \$30-50B
- Multiple: 20-25x
- **Milestone:** IPO or strategic acquisition

Year 5+:

- Revenue: \$2-3B ARR
 - Market cap: \$40-60B (public market)
 - **Outcome:** The defining AI infrastructure company for the physical world
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The Path to IPO

Why we go public (vs stay private):

Pros of IPO:

-  Liquidity for employees/investors
-  Currency for acquisitions (stock)
-  Brand credibility (public company = enterprise trust)
-  Access to unlimited capital (public markets)

Cons of staying private:

-  Reporting requirements (quarterly earnings pressure)
-  Short-term thinking from public markets
-  Competitive intel exposed (revenue, margins public)

Our likely path:

- Stay private through Year 3 (2027)
- IPO in Year 4-5 (2028-2029) when revenue >\$1.5B
- Dual-class share structure (retain founder control)

IPO Comparable:

- Snowflake IPO (2020): \$3.4B raised at \$33B valuation (12x ARR)
 - Databricks (expected 2025): Likely \$50B+ at IPO
 - **Skanda (2028-29):** \$40-60B at IPO (targeting \$2B revenue)
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THE GO-TO-MARKET STRATEGY

How we acquire customers across all four revenue layers

Layer 0 (API): Viral + Self-Service

Top of Funnel (Awareness):

- Publish research (Nature, Science, NeurIPS)
- Open-source tools (microstructure visualization, data processing)
- Content marketing (blog, YouTube tutorials, webinars)
- Academic partnerships (free API for universities)

Conversion (Free → Paid):

- Free tier: 100 predictions/month (hook them in)
- Usage notifications: "You've used 90/100 free predictions this month. Upgrade for \$500?"
- Product-led growth: No sales calls for <\$5k/month deals

Expansion (Developer → Professional → Enterprise):

- In-app upgrade prompts
- Usage analytics: "You could save 30% by upgrading to Professional tier"
- White-glove onboarding for Enterprise (>\$50k/year)

Target CAC:

- Developer tier: \$50-100 (digital marketing)
- Professional tier: \$500-1,000 (inside sales)
- Enterprise tier: \$20-50k (field sales)

Target LTV:CAC: 10:1 or better (standard for great SaaS)

Layer 1 (NRE): Enterprise Sales

Ideal Customer Profile:

- Revenue >\$1B (can afford \$5M partnership)
- Active R&D budget >\$50M/year

- Pain: "We're 3-5 years behind on next-gen chemistry"

Target Accounts (Tier 1 Priority):**Automotive OEMs:**

- BMW, Mercedes, Tesla, Tata, BYD
- Pain: Dependent on external suppliers (CATL, LG)
- Value prop: "Own your battery IP, break supplier lock-in"

Battery Manufacturers:

- CATL, LG Energy, Panasonic, Samsung SDI
- Pain: R&D takes too long, competitors catching up
- Value prop: "2x R&D speed, 50% cost reduction"

Next-Gen Startups:

- QuantumScape, Factorial, Solid Power
- Pain: Burning cash on trial-and-error
- Value prop: "Get to production faster, conserve runway"

Defense/Aerospace:

- Lockheed, Northrop, Safran, ISRO
- Pain: Extreme requirements (-40°C to +85°C operation)
- Value prop: "Design for edge cases impossible with standard methods"

Sales Process:

1. Intro call (15 min pitch)
2. Technical deep-dive (show API demo + case study)
3. Pilot proposal (\$500k for 3 months)
4. Pilot execution (prove ROI)
5. Full partnership (\$3-7M)

Sales Cycle: 6-9 months (enterprise standard)

Team Structure:

- 1 Head of Enterprise Sales

- 3-5 Account Executives (each owns 20-30 accounts)
 - 2-3 Sales Engineers (technical pre-sales)
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Layer 2 (SaaS): Partner Channel + Direct

Customer Profile:

- Existing NRE partners (upsell after IP phase)
- Mid-tier manufacturers (don't need custom chemistry, just want yield optimization)

Sales Motion:

1. "You've designed the chemistry with us (Layer 1)"
2. "Now let's ensure your factory actually produces it correctly (Layer 2)"
3. Offer bundled discount: NRE + SaaS = 15% off

Partner Channel:

- Equipment OEMs (Bühler, Hibar, PNE, Sovema)
- System integrators (Siemens, Dassault Systèmes)
- They bundle our software with their equipment

Why partners sell for us:

- Makes their equipment "smarter" (competitive differentiation)
 - Recurring revenue share (20% of SaaS fee)
 - Stickier customers (harder to switch equipment vendors)
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Layer 3 (Royalties): IP Licensing

This layer is passive — customers come to us

Scenario A: NRE partners reach end of exclusivity

- BMW's 3-year exclusivity on Silicon design expires
- We offer to other OEMs: "License BMW's proven design for \$2M + \$0.75/kWh"
- BMW can extend exclusivity by paying higher royalty rate

Scenario B: API users scale to production

- Startup used our API to design novel chemistry
- They reach 10 GWh/year production (royalty threshold)
- Automatic royalty via terms of service

Scenario C: Inbound licensing

- We publish IP portfolio (public database of our designs)
- Manufacturers approach us: "We want to license your solid-state design"
- Standard terms: \$5M upfront + \$1.50/kWh

Enforcement:

- Contractual audits (right to inspect production data)
- Partnership with battery analytics firms (track global production)
- Legal team for IP enforcement (China will try to copy)

THE CAPITAL ALLOCATION

How we deploy the \$20M to maximum effect

50% — Product & Engineering (\$10M)

The Investment:

- **Platform Engineers:** 12-15 senior engineers (\$150-200k/year total comp)
 - API infrastructure (4-5 engineers)
 - Edge deployment / federated learning (3-4 engineers)
 - DevOps / security / reliability (2-3 engineers)
 - Data engineering / ETL (2-3 engineers)
- **ML Researchers:** 5-8 PhDs (\$120-180k/year)
 - Foundation model development (2-3 researchers)

- Transfer learning optimization (2 researchers)
- Computer vision for SEM analysis (1-2 researchers)
- Physics simulation experts (1-2 researchers)
- **Product Managers:** 2-3 (\$100-150k/year)
 - API product (1 PM)
 - Dashboard/self-service (1 PM)
 - Enterprise features (1 PM)

Why 50% here:

- Product is the moat
- Best engineers = best platform = winner-take-most
- AI infrastructure is engineering-heavy (like Databricks, Snowflake)

Expected Output:

- Production-grade API (99.9% uptime)
- Self-service platform (1,000+ users)
- Foundation model v2 (multi-chemistry)

25% — Data Generation (\$5M)

The Investment:

- **Lab Equipment:** \$500k one-time
 - Gloveboxes, cyclers, mixers, ovens (no autonomous lab yet)
- **Materials Scientists:** 6-8 people (\$80-120k/year)
 - 2 per chemistry (Silicon, cathodes, hydrogen, sodium)
 - 1 senior scientist (owns quality control)
- **Materials & Consumables:** \$150k/month
 - Silicon, graphite, binders, solvents, electrolytes
 - Coin cell hardware
- **Characterization:** \$1M/year
 - SEM/XRD partner lab access

- Eventually buy our own (Year 2)

Why 25% here:

- Data is the defensible asset
- Manual lab is 5x cheaper than autonomous (and faster for first 1,000 samples)
- Quality over quantity (300 perfect samples > 3,000 noisy samples)

Expected Output:

- 1,000+ multimodal samples across 4 chemistries
 - Largest proprietary mesoscale dataset globally
-

15% — Go-to-Market (\$3M)**The Investment:**

- **Forward Deployed Engineers:** 5-8 people (\$150-200k/year fully loaded)
 - Embed at Tier-1 partner sites
 - Custom integration, training, support
- **Enterprise Sales:** 4-6 people (\$120-180k/year + commission)
 - 1 Head of Sales
 - 3-4 Account Executives
 - 1-2 Sales Engineers
- **Developer Relations:** 2-3 people (\$100-120k/year)
 - API docs, tutorials, webinars
 - Community building (Discord, Stack Overflow)
 - Conference talks (NeurIPS, MRS, Electrochemical Society)
- **Marketing:** \$300k/year
 - Content creation (blog, case studies)
 - Paid ads (Google, LinkedIn)
 - Events (booths at industry conferences)

Why 15% here:

- API is mostly self-service (low CAC)
- Enterprise deals fund themselves (high LTV)
- Developer advocacy = viral growth

Expected Output:

- 8-12 enterprise partnerships (\$40-60M revenue)
 - 100-200 API customers (\$5-10M revenue)
 - 20-30 production line deployments (\$6-9M revenue)
-

10% — IP & Infrastructure (\$2M)

The Investment:

- **Patents:** \$1M (50-100 patents × \$10-20k each)
 - Microstructure designs
 - Process recipes
 - ML architecture innovations
- **Legal:** \$500k
 - Partnership contracts
 - Licensing agreements
 - Corporate governance (Series A, B compliance)
- **Cloud Infrastructure:** \$500k
 - AWS/Azure credits (API hosting)
 - GPU clusters for training
 - Data storage & CDN

Why 10% here:

- IP is the long-term moat
- Patents take 3-5 years to issue (start early)
- Infrastructure is cheap (we get government compute for free via IndiaAI)

Expected Output:

- 50-100 patent applications filed

- Bulletproof IP position (defensible for 15-20 years)

The Capital Efficiency Argument

Our \$20M achieves what \$80-100M achieves in Silicon Valley:

EXPENSE	SILICON VALLEY	INDIA (SKANDA)	SAVINGS
Senior ML Engineer	\$300k/year	\$150k/year	50%
Platform Engineer	\$250k/year	\$120k/year	52%
Lab Technician	\$80k/year	\$30k/year	62%
Office/Infrastructure	\$5k/person/month	\$1k/person/month	80%
Total (80 people, 18mo)	\$80-100M	\$20M	75%

Plus government subsidies:

- H100 GPU cluster access (saves \$5-8M/year)
- Lab equipment partnerships (saves \$2-3M)
- R&D tax credits (25% of spend)

Result: We reach \$200M ARR for 1/4 the capital of a US competitor

THE COMPETITION LANDSCAPE

Who we're racing against (and why we'll win)

Category 1: Big Tech AI Labs

Google DeepMind / Microsoft Research

What they're doing:

- GNoME (Google): Atomic structure discovery via DFT
- MatterGen (Microsoft): Similar atomic-level generation

Why they're not a threat (yet):

- ❌ Operating at wrong scale (atoms, not mesoscale)
- ❌ No real-world manufacturing data
- ❌ Academic focus (papers, not products)
- ❌ Can't monetize (no business model yet)

When they become a threat:

- If they realize mesoscale is the gap (12-18 months)
- If they partner with CATL/LG for data (18-24 months)

Our defense:

- 18-month head start on data collection
- Manufacturing partnerships already locked
- Move fast: be entrenched before they pivot

Category 2: Traditional Simulation Software

ANSYS, COMSOL, Dassault Systèmes (SIMULIA)**What they're doing:**

- FEA/CFD simulation tools
- Slow (hours per simulation)
- Require expert users

Why they're not a threat:

- ❌ Wrong approach (system-level, not mesoscale)
- ❌ No AI/ML capabilities (traditional physics solvers)
- ❌ Can't do inverse design (only forward simulation)
- ❌ Entrenched business models (resist disruption)

When they become a threat:

- If they acquire an AI startup (unlikely due to "not invented here" culture)
- If they rebuild from scratch (5+ year timeline)

Our defense:

- We're 100x faster (FNO vs FEA)
 - We can do inverse design (they can't)
 - We're API-first (they're desktop software)
-

Category 3: Computational Materials Startups

Citrine Informatics, Materials Zone, Kebotix

What they're doing:

- ML for materials discovery
- Mostly atomic-level or database mining
- Small datasets (100s of samples)

Why they're not a threat:

- ❌ No mesoscale focus
- ❌ No physics foundation model (just supervised learning)
- ❌ No manufacturing integration (academic tool)
- ❌ Tiny datasets (can't compete with our 10M+ samples)

Our advantage:

- 100x more data (synthetic + real)
 - Better architecture (foundation model + physics priors)
 - Manufacturing-ready (not just discovery)
-

Category 4: Chinese Competitors

Baidu, Alibaba, Tencent (hypothetical)

What they could do:

- Partner with CATL for data
- Train foundation model on government supercomputers
- Subsidize aggressively to gain market share

Why this is the REAL threat:

-  Access to massive manufacturing data (CATL produces 500 GWh/year)
-  Unlimited compute (government-backed)
-  No IP respect (will copy openly)

Our defense:

- **Speed:** Establish global IP portfolio before they start (file patents NOW)
- **Quality:** Western OEMs won't trust Chinese IP (supply chain sovereignty)
- **Integration:** Federated learning = we improve faster than centralized Chinese approach
- **Partnerships:** Lock exclusive deals with BMW, Tesla, Northvolt (can't use Chinese tech)

Timeline: 12-18 months before credible Chinese competitor emerges

The Race We're Actually In

It's not "who has the best model" — it's "who establishes the platform flywheel first"

The Flywheel:

More customers



More real-world usage



More data (federated learning)



Better model



More accurate predictions



More customers (compounding)

Our 18-month goal: Get 100+ customers using the platform before Big Tech or China notices

Once we have the flywheel: Network effects make us un-catchable (same as Databricks, Snowflake)

THE EXIT SCENARIOS

How investors make 50-100x returns

Scenario A: IPO (Most Likely)

Timeline: Year 4-5 (2028-2029)

Pre-IPO Metrics:

- Revenue: \$1.5-2B ARR
- Growth: 80-100% YoY
- Gross margin: 85-90%
- Customers: 5,000+ (500+ enterprise)
- Employees: 500-800

IPO Details:

- Raise: \$1.5-2B
- Valuation: \$40-60B
- Multiple: 25-30x revenue (AI infrastructure premium)
- Exchanges: NYSE or NASDAQ
- Dual-class shares (founders retain control)

Comparable: Snowflake IPO (2020) raised \$3.4B at \$33B valuation (12x revenue)

Investor Returns:

- Series A investor (\$5M at \$50M pre-money): 800-1,200x
- Series B investor (\$50M at \$1B pre-money): 40-60x

- Series C investor (\$200M at \$5B pre-money): 8-12x
-

Scenario B: Strategic Acquisition (Alternative)

Potential Acquirers:

Microsoft (\$3T market cap):

- Rationale: Integrate into Azure (cloud-native materials platform)
- Price: \$30-50B (2-3% of market cap)
- Precedent: GitHub (\$7.5B), Nuance (\$20B), Activision (\$69B)

Amazon (\$1.7T market cap):

- Rationale: AWS for manufacturing (expand beyond digital)
- Price: \$25-40B
- Precedent: Whole Foods (\$13.7B), MGM (\$8.5B), iRobot (\$1.7B)

NVIDIA (\$3T market cap):

- Rationale: Omniverse + materials (digital twin for physical world)
- Price: \$30-60B
- Precedent: Mellanox (\$7B), Arm (attempted \$40B)

Tesla (\$800B market cap):

- Rationale: Vertical integration (own battery IP platform)
- Price: \$20-35B
- Precedent: SolarCity (\$2.6B), DeepScale (\$undisclosed)

CATL / LG Energy (Battery Incumbents):

- Rationale: Defensive (prevent competitors from accessing platform)
- Price: \$15-25B
- Less likely (anti-trust concerns)

Timeline: Year 3-5 (after we're clearly winning)

Why we'd consider it: If acquirer can accelerate global deployment faster than we can alone

Scenario C: Stay Private Longer (Unlikely but Possible)

If capital markets are unfavorable:

- Raise large late-stage rounds (\$500M-1B)
- Delay IPO until \$3-5B revenue
- Go public at \$80-120B valuation

Precedent: Databricks (still private at \$43B, 10 years old)

Downside: Liquidity event delayed for employees/early investors

THE RISKS & MITIGATION

What could go wrong (and how we're preparing)

Risk 1: Sim-to-Real Transfer Fails

The risk: Model trained on synthetic data doesn't predict real-world performance

Likelihood: Medium (this is the classic deeptech failure mode)

Impact: High (undermines entire platform thesis)

Mitigation:

-  Domain adaptation (already built into architecture)
-  Generate 300+ real samples in first 6 months (validate early)
-  Conservative claims (don't promise 100% accuracy, aim for 85-90%)
-  Continuous learning (model improves as we collect more real data)

Validation checkpoint: Month 6 — if $R^2 < 80\%$ on real data, pivot strategy

Risk 2: Big Tech Copies Us

The risk: Google/Microsoft train mesoscale model and bundle into cloud platform

Likelihood: Medium-High (they will eventually notice)

Impact: High (could commoditize our platform)

Mitigation:

-  **Speed:** Lock 100+ customers before Big Tech ships (18-month head start)
-  **Data moat:** Our real-world dataset is un-replicable (takes years to collect)
-  **Network effects:** Federated learning = we improve faster than centralized approach
-  **Domain expertise:** We have materials scientists, they don't (harder to copy than pure software)

When this happens: Likely 2027-2028 (by then we should have platform lock-in)

Risk 3: Market Adoption Slower Than Expected

The risk: Manufacturers are conservative, don't trust AI

Likelihood: Medium (industrial sales cycles are slow)

Impact: Medium (delays revenue, not fatal)

Mitigation:

-  **Proof points:** Publish Nature paper (credibility with technical buyers)
-  **Reference customers:** BMW/Tesla using us = validation for others
-  **Risk reversal:** Offer "success-based pricing" (pay only if it works)
-  **Freemium API:** Let engineers experiment risk-free

Contingency: If enterprise is slow, focus on API growth (developer-led, faster adoption)

Risk 4: Chinese IP Theft

The risk: CATL or Chinese competitors copy our designs without paying royalties

Likelihood: High (China's track record on IP)

Impact: Medium (limits royalty revenue in China)

Mitigation:

-  **Geographic focus:** Target US/EU/Japan/India first (strong IP enforcement)
-  **Trade secrets:** Keep some key innovations as trade secrets (not patentable)
-  **Operational security:** Federated learning = data never centralizes in one place
-  **Strategic alliances:** Partner with governments (IndiaAI, US DOE) for geopolitical support

Acceptance: We may never monetize China fully (but 70% of global EV production is outside China)

Risk 5: Key Person Dependency

The risk: Founders/key scientists leave or become unavailable

Likelihood: Low (but high impact)

Impact: High (IP is in people's heads during early phase)

Mitigation:

-  **Documentation:** Codify all scientific knowledge in papers, repos, wikis
 -  **Team redundancy:** Two experts per critical area (not single points of failure)
 -  **Retention:** Competitive comp, meaningful equity, inspiring mission
 -  **Succession plan:** Identify internal candidates for leadership roles
-

Risk 6: Model Bias or Hallucination

The risk: AI confidently predicts wrong answer, customer builds it, it fails catastrophically

Likelihood: Medium (all ML models have failure modes)

Impact: High (liability, reputation damage, customer churn)

Mitigation:

-  **Confidence scores:** Always output uncertainty estimates
-  **Human-in-loop:** Require expert review for mission-critical designs
-  **Validation protocols:** "AI generates, human validates, lab confirms"
-  **Insurance:** Carry errors & omissions insurance (\$10M+ coverage)
-  **Transparency:** Show customers the training data distribution (when to trust vs not trust)

Legal protection: Terms of service clarify we provide "decision support" not "autonomous decisions"

THE TEAM WE'RE BUILDING

Who we need to win

Founding Team (Current)

Technical Leadership:

- Founders with PhDs in Materials Science / AI
- Published in top-tier venues (Nature, NeurIPS, Science)
- IndiaAI Mission backing (government validation)

What we've proven:

-  Built foundation model (10M+ synthetic samples)
 -  Achieved 99% R^2 on synthetic data
 -  Developed diffusion-based inverse model
 -  Secured government compute & lab infrastructure
-

Key Hires (Next 18 Months)

Priority 1: VP Engineering (Month 0-3)

- Experience: Built API platforms at scale (100k+ RPS)

- Background: Google, Meta, Databricks, Snowflake
- Skills: Distributed systems, ML infrastructure, team building
- Comp: \$300-500k + 1-2% equity

Priority 2: Head of Enterprise Sales (Month 0-6)

- Experience: Sold \$5-10M deals to automotive/battery OEMs
- Background: Siemens, Dassault, ANSYS, or battery supplier
- Skills: C-level relationships, consultative selling
- Comp: \$200-300k base + commission + 0.5-1% equity

Priority 3: Chief Scientist (Month 6-12)

- Experience: 15-20 years in battery R&D, published 50+ papers
- Background: CATL, LG, Panasonic, or top academic lab (MIT, Stanford)
- Skills: Technical credibility with customers, IP strategy
- Comp: \$250-400k + 0.5-1% equity

Priority 4: Head of Product (Month 6-12)

- Experience: 0→1 API products, developer tools
- Background: Stripe, Twilio, AWS, or similar
- Skills: Product-led growth, developer experience
- Comp: \$200-350k + 0.5-1% equity

Team Scaling Trajectory

PHASE	HEADCOUNT	BREAKDOWN
Month 6	20-25	10 eng, 5 sales/FDEs, 3 scientists, 2 ops
Month 12	40-50	20 eng, 10 sales/FDEs, 8 scientists, 5 ops, 5 product
Month 18	80-120	40 eng, 20 sales/FDEs, 15 scientists, 10 ops, 10 product, 5 exec

THE ASK

What we're raising and why

Series A: \$20M

Target Close: Month 0-3 (Q2 2025)

Use of Funds:

- 50% Product & Engineering
- 25% Data Generation
- 15% Go-to-Market
- 10% IP & Infrastructure

Milestones to be achieved:

- Month 6: Silicon validated ($>90\%$ R^2 on real data)
- Month 12: Platform launched, \$52-60M cumulative revenue
- Month 18: \$110-135M cumulative revenue (\$50-60M ARR), ready for Series B

Valuation: \$80-120M pre-money

Investor Returns (at IPO in Year 4-5):

- Entry: \$5M at \$100M post-money (5% ownership)
- Exit: 5% of \$50B = \$2.5B (pre-dilution)
- **Return: 100-200x over 4-5 years (post-dilution)**

Why This Round Sets Us Up to Win

\$20M is the perfect amount:

- **✗** Too little (\$5-10M): Can't hire fast enough, competitors catch up
- **✗** Too much (\$50M+): Dilutes founders, creates pressure for quick exits

-  Just right: 18-24 months runway to \$110-135M cumulative revenue → Series B at \$3-5B valuation

What we're NOT raising for:

-  Autonomous lab (\$5M, too slow for Phase 1)
-  Building batteries ourselves (capital intensive, wrong focus)
-  Geographic expansion beyond Munich/Tokyo (premature)

What we ARE raising for:

-  Best-in-class product team (winner-take-most market)
 -  High-quality data (defensible moat)
 -  Strategic partnerships (enterprise validation)
-

Investor Profile (Who We Want)

Tier 1 (Lead Investor):

- AI-focused funds: Andreessen Horowitz (a16z), Sequoia, Greylock
- Deep tech specialists: Lux Capital, DCVC, Breakthrough Energy Ventures
- Check size: \$10-15M
- Value-add: Network to Big Tech acquirers, AI talent recruitment

Tier 2 (Strategic Investors):

- Corporate VCs: BMW i Ventures, LG Technology Ventures, Energy Impact Partners
- Check size: \$2-5M
- Value-add: Customer intros, pilot partnerships, industry expertise

Tier 3 (Specialist Angels):

- Technical founders: Ex-Googlers, Databricks alumni, battery industry veterans
 - Check size: \$100k-500k
 - Value-add: Hiring intros, technical advice, credibility
-

APPENDIX: THE NUMBERS

Detailed financial projections

Revenue Build (18-Month Projection)

MONTH	API ARR	NRE (CUMULATIVE)	SAAS ARR	IP ARR	TOTAL CUMULATIVE	ARR RUN- RATE
0	\$0	\$0	\$0	\$0	\$0	\$0
3	\$0.5M	\$5M	\$0	\$0	\$5.5M	\$0.5M
6	\$2M	\$12M	\$0.5M	\$0	\$14.5M	\$2.5M
9	\$6M	\$22M	\$2M	\$0.5M	\$30.5M	\$8.5M
12	\$12M	\$35M	\$4M	\$1M	\$52M	\$17M
15	\$22M	\$50M	\$7M	\$2M	\$81M	\$31M
18	\$35M	\$65M	\$12M	\$4M	\$116M	\$51M

Note: NRE is one-time (shown as cumulative). ARR Run-Rate = API + SaaS + IP (recurring revenue only)

Customer Acquisition (18-Month Projection)

SEGMENT	MONTH 6	MONTH 12	MONTH 18	ARPU	CAC	LTV:CAC
API Free	200	500	2,000	\$0	\$0	∞
API Developer	20	100	500	\$6k/yr	\$100	60:1
API Professional	5	50	200	\$60k/yr	\$1k	60:1
API Enterprise	0	5	20	\$600k/yr	\$50k	12:1
NRE Partnerships	3	8	15	\$5M	\$200k	25:1

Factory SaaS	5	15	30	\$400k/yr	\$100k	4:1
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Unit Economics (Mature State — Year 3+)

API Business:

- COGS: \$0.001 per prediction (compute)
- Gross Margin: 99%
- CAC: \$500 (blended)
- LTV: \$30k (3-year avg customer lifespan × \$10k/year)
- LTV:CAC: 60:1

Enterprise/NRE:

- COGS: 30% (labor, lab costs)
- Gross Margin: 70%
- CAC: \$100k
- LTV: \$5M (partnership) or \$2M (3-year SaaS contract)
- LTV:CAC: 20-50:1

Blended:

- Gross Margin: 85-90%
- LTV:CAC: 30:1+

Hiring Plan

ROLE	MONTH 0	MONTH 6	MONTH 12	MONTH 18
Engineering	5	8	18	35
Science/Lab	3	5	8	15
Sales/FDEs	2	4	8	18
Product	1	1	4	8

Ops/Admin	2	2	3	8
Exec	3	3	5	8
Total	16	23	46	92

Burn Rate & Runway

PHASE	MONTHLY BURN	PERIOD SPEND	PERIOD CASH IN	CASH AT END
Month 0-6	\$1M	\$6M	\$3M (NRE deposits)	\$17M
Month 6-12	\$1.5M	\$9M	\$25M (NRE + API)	\$33M
Month 12-18	\$2M	\$12M	\$45M (all streams)	\$66M

Interpretation: Cash-flow positive by Month 9-10. By Month 18, we have **\$66M in the bank** from revenue WITHOUT raising Series B.

This gives us optionality: We raise Series B from a position of strength, not desperation — we choose the right partner, not the fastest money.

THE CLOSING ARGUMENT

Why Skanda is the defining infrastructure company of the 2020s

We are at an inflection point in human civilization.

The AI revolution has conquered the digital world. ChatGPT, Midjourney, Claude — they generate text, images, code. But they stop at the screen.

The physical world is still stuck in the 20th century. Developing a new material takes 5-10 years and \$10M+. We're running out of time for the energy transition, for climate tech, for advanced manufacturing.

Skanda bridges the gap.

We're not building another LLM. We're building the **AI that builds the physical world.**

- Need a battery that charges in 5 minutes? → We generate the microstructure
- Need cement that sequesters CO₂? → We optimize the formulation
- Need an alloy that withstands Mach 5? → We design the grain structure

This is not science fiction. This is operational in 18 months.

We have:

- ☒ The science (10M+ simulations, 99% R² on synthetic data)
- ☒ The team (IndiaAI-backed, published in Nature)
- ☒ The capital efficiency (5x cheaper than Silicon Valley)
- ☒ The market timing (energy transition = secular tailwind)

What we need:

- ☒ \$20M to execute the 18-month sprint
- ☒ Partners who believe the physical world deserves the same AI revolution as the digital world

The outcome:

- 18 months: \$3-5B valuation (platform proven, \$50-60M ARR)
- 5 years: \$40-60B valuation (multi-industry dominance)
- 10 years: The defining infrastructure company for the physical world

The choice is simple:

Invest in another SaaS tool that improves margins by 5%?

Or invest in the platform that **accelerates human civilization** by 10 years?

We're building the operating system for physical matter.

Join us.

END OF DOCUMENT